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1 最短路徑

1.1 basic

```
1 // c++ code
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 int main() {
6     // test comment
7     cout << "test string\n";
8 }
```

2 圖論

2.1 凸包問題

```
1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return
5         (x<other.x) || (x==other.x && y<other.y);
6     };
7 };
8
9 struct ConvexHull{
10     vector<node> vec;
11     int n;
12     void init(vector<node> &v){
13         vec=v;
14         n=v.size();
15     }
16
17     int cross(node a,node o,node b){
18         return
19         (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
20     }
21
22     vector<int> point;
23     vector<int> area;
24
25     void build(){
26         point=vector<int>(n+1);
27         area=vector<int>(n+1);
28         vector<node> up,down;
29         int upsum=0;
30         int downsum=0;
31         for(int i=0;i<n;i++){
32             while(up.size()>=2 &&
33                 cross(up[up.size()-2],
34                     up[up.size()-1],
35                     vec[i])<=0){
36                 upsum-=cross(up[up.size()-2],
37                             {0,0},up[up.size()-1]);
38                 up.pop_back();
39             }
40             up.push_back(vec[i]);
41             if(up.size()>1)
42                 upsum+=cross(up[up.size()-2],
```

```
43         {0,0},
44         up[up.size()-1]);
45
46         while(down.size()>=2 &&
47             cross(down[down.size()-2],
48                 down[down.size()-1],
49                 vec[i])>=0){
50             downsum-=cross(down[down.size()-2],
51                             {0,0},down[down.size()-1]);
52             down.pop_back();
53         }
54         down.push_back(vec[i]);
55         if(down.size()>1)
56             downsum+=cross(down[down.size()-2],
57                             {0,0},down[down.size()-1]);
58
59         area[i+1]=abs(upsum-downsum);
60
61         if(i+1==1) point[i+1]=1;
62         else point[i+1]=up.size()+down.size()-2;
63     }
64 }
65
66 double getarea(int p){
67     return area[p]/2.0;
68 }
69
70 int getnum(int p){
71     return point[p];
72 }
73 };
```

3 數學

3.1 公式

• 中文測試

$$\cdot \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$